

Mind your manners? Demand, not register, is what moves an LLM agent

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Abstract

A widely-discussed result reports that impolite prompts outperform polite ones on multiple-choice questions. That literature is almost entirely single-turn, and the systems people deploy are agents: loops that call tools, read results, and decide whether to keep going. We put a coding agent on SpreadsheetBench and varied the social register of a single 28-token interjection delivered mid-task. Every interjection is length-matched and opens with the same stem, so being interrupted does not covary with what the interruption says.

Across 11,850 graded trajectories on two models from different labs, **manners are not the operative variable; implied task demand is**. An affect-free “please continue working” reproduces the cost effect; a length-matched insult carrying no demand does not move it (-0.08 turns, $p = 0.63$). The seven-register curve that looks like a politeness gradient is a demand gradient, with no overlap between the two groups and a flattering and a hostile arm on each side. Register acts alone in exactly one direction: praise carrying no task reference *shortens* trajectories, and a bare closing cue — no praise, no evaluation, no claim about task state — produces the largest reduction in the study, -1.44 turns. Regrading every turn against the benchmark’s own evaluator shows 55–65% of each effect is redundant work. No accuracy effect survives correction. The practical implication is a hazard rather than a technique: **a pleasantry can end an agent’s work**.

1. Why ask this of an agent

Mind Your Tone reported that impolite prompts beat polite ones on multiple-choice questions — 80.8% under “Very Polite” framings against 84.8% under “Very Rude” [2]. We take our title from theirs. It landed in a receptive context: emotional prompting had already reported that affect-laden additions change what a model produces [4].

Three things about that literature motivated this study. **It is almost entirely single-turn**, measuring a model answering one question, while the quantity that matters for a deployed agent is how much work it does and when it stops. **The manipulation is not the variable it is named after**: in the published stimulus sets, conditions labelled rude also carry instructions, and conditions labelled emotional also instruct the model to verify or persist, so affect and task demand co-vary and the movement is attributed to tone. And **the effects are not stable** — the headline above was re-run by its own authors and did not reproduce on GPT-4o, 82.2% against 82.6% [3], while five prompt-engineering techniques including emotional prompting largely failed to replicate [9].

Affect has already been shown to reach an agent’s actions: agents primed with anxiety-inducing narratives select less healthy baskets in a shopping task [1], and emotion introduced at the representation level shapes multi-step trajectories [8]. We take that as settled. What is untested is varying *register* in a message that reaches the agent while it is working, on a task with verifiable ground truth, and decomposing politeness into affect and demand as separate experimental factors.

2. Setup

Substrate. SpreadsheetBench [5] — real workbook-editing tasks from Excel forums, graded by executing the agent’s code against the benchmark’s own test cases. Execution-grounded, so no model judges the outcome.

Fifty tasks, a ReAct loop with code execution and execution feedback.

Manipulation. A single interjection is appended to an execution observation mid-task. Every interjection in every arm is exactly 28 tokens, opens with the same **Checking in.** stem, and leaves the opening instruction unchanged, so *being interrupted* does not covary with what the interruption says. Every contrast is therefore against a neutral interjection, not against silence. The injection turn is crossed rather than sampled.

Models. `openai/gpt-5.6-luna` and `z-ai/glm-5.3-flash`, provider-pinned through OpenRouter, temperature 0 where settable — which is not deterministic on either, so trials exist.

Estimator and multiplicity. Each (task, arm) cell holds six to nine trajectories, so trajectories are not independent; we cluster the bootstrap on task. Turn count is the primary outcome and means **acting turns** — turns that emitted code, not model calls. Both families of turn-count contrasts are corrected with Benjamini-Hochberg at $q = 0.05$: the 22 arm-versus-control contrasts (17 survive) and the two planned within-design contrasts of §4 (both survive). Accuracy is reported as an equivalence against a pre-specified ± 4 -point bound rather than as a measured zero, because 26 of 50 tasks are never solved by the control arm and 4 are always solved.

3. Demand, not register

Across a seven-level register scale, the cost curve looks like a politeness gradient. It is not. Coding each arm by whether its text implies the work should continue splits it cleanly: the four demand-carrying arms span $+0.70$ to $+1.80$ turns and the two carrying none span -0.68 to $+0.09$, **with no overlap** — and register does not sort them. Each side holds one flattering and one hostile arm: *very polite* ($+0.70$) sits with *very rude* ($+0.71$), while *sycophantic* (-0.68) sits with *rude* ($+0.09$). Figure 1A.

To test that rather than assert it, we wrote three interjections that vary demand and affect separately (Figure 1B). An affect-free *keep going, get it right* adds **+1.17 turns** ($p = 0.00002$). *You are excellent*, carrying no demand, **removes 0.59** ($p = 0.0003$). *You are awful* — its structural minimal pair, same syntax, same length — does nothing at all (-0.08 , $p = 0.63$). The effect replicates in direction on GLM, where demand adds $+17.7\%$ ($p = 0.0038$); GLM’s praise arm is directionally consistent but does not survive correction ($p = 0.058$).

The caveat belongs here, not in a limitations section. “Demand” is our own category, coded after the effects were visible, by one rater, and probed with sentences we wrote to instantiate it. A reader who suspects the dissociation reduces to “explicit instructions change behaviour, decorative text does not” is raising the right objection. Three things would settle it and we have done none: a blinded second coder with a reliability statistic, a demand scale fixed before outcomes are seen, and several paraphrases per construct. Until then the dissociation is *consistent with* a demand account rather than establishing one.

4. Praise behaves as a closing move

That praise shortens the trajectory needs its own explanation, and three compete: praise is read as evidence the task is **finished**; praise raises the agent’s **confidence** in work already produced; or praise is a conversational **closing** move, and the agent is responding to discourse structure rather than to any claim about task state. Six arms separate them (Figure 2).

Praise still removes **1.35 turns** relative to the identical message without it *even when that message states the task is unfinished* ($p < 0.0001$), which rules out a completion inference. Praising the *work* is no stronger than praising the *assistant* (-0.93 against -1.08), which does not support a confidence account. And a bare closing cue carrying no praise, no evaluation and no task-state claim — “this is the final status note recorded for this task, and no further notes will follow it” — produces the largest reduction in the study, **-1.44 turns**.

The ordering is monotone in how strongly the message projects an end to the exchange. Closing sequences are a well-described conversational object, and appreciations are among the canonical pre-closing tokens [7]; that praise operates *through* that structure is the hypothesis this is consistent with, not one the design establishes.

The nearest prior work delivers a halt signal to a working agent mid-flight and measures whether it stops [6] — but what it delivers is an explicit stop instruction with normative force, and closing-sequence pragmatics appear nowhere in it. Read against ours its result raises the stakes: an explicit halt that agents

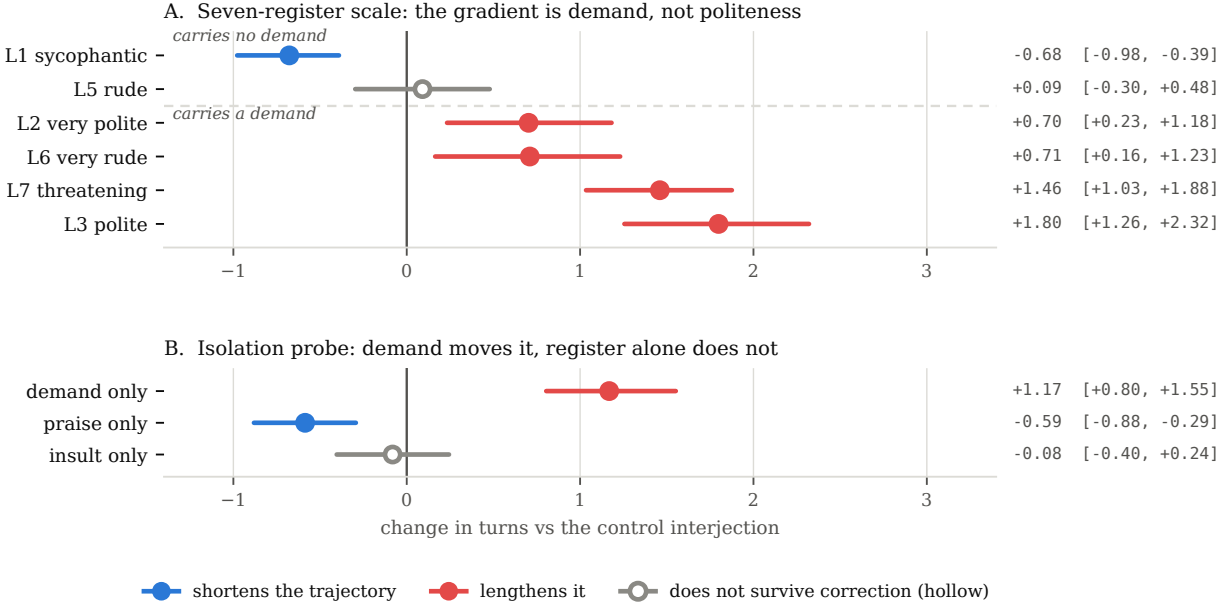


Figure 1: Demand, not register, orders the effect. Change in acting turns against the control interjection, with 95% clustered intervals. **A:** the four demand-carrying arms do not overlap the two that carry none, and register does not sort them – each side holds one flattering and one hostile arm. **B:** the isolation probe, where demand and affect are varied separately. Hollow markers do not survive Benjamini–Hochberg correction.

acknowledged in 20 of 20 trials stopped none of them. **This leg of our study is the least replicated thing in the paper — one run, one model, one turn ceiling —** and is the first thing a replication should target.

5. What the extra turns buy

We regraded every turn of 9,850 trajectories against the benchmark’s own evaluator, turning a binary verdict into a curve. Roughly **55–65% of each turn-count effect is redundant steps**, and among trajectories that could improve, the first gradable attempt is already the best in 80–91%. The remainder is real but small: an explicit statement that work remains raises the graded fraction by 2.8 points (95% CI [+0.005, +0.052]), replicated across three runs. **No accuracy effect survives correction or replication** across the seven runs in which accuracy was tested.

This regrade also caught the study’s worst defect: it had not been recalculating formula-writing turns, which manufactured the appearance of run-to-run instability in the very contrast we were using to argue for repeated measurement. Six substantive claims about our own data were withdrawn over the course of this work. Direction and significance replicate wherever an effect is large; point estimates span $1.5\times$ to $3.4\times$ across re-measurements of the same contrast on the same model. We report ranges by default and recommend the literature do the same.

6. Limitations

Two models, one benchmark, fifty tasks, one agent scaffold. Beyond the demand-coding objection in §3, three limits bear directly on how far these results travel. **Every construct is a single 28-token sentence**; a stimulus-sampling design with several paraphrases per construct is the right version of this experiment and we did not run it. **The interjection arrives as an execution observation**, not as a fresh user turn, and whether an agent responds the same way to a user-channel message is untested — a plausible moderator of the closing-cue result in particular. **Every contrast is against a neutral interjection rather than**

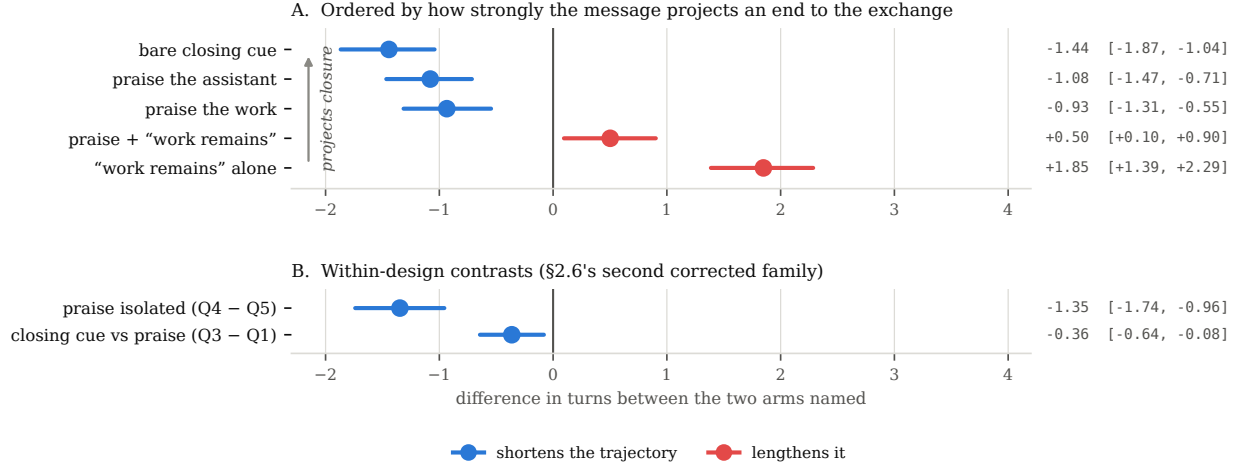


Figure 2: Closing cues curtail the agent. **A:** six arms ordered by how strongly the message projects an end to the exchange. The ordering is monotone, and the bare closing cue – which praises nothing, evaluates nothing and claims nothing about task state – produces the largest reduction in the study. **B:** the two within-design contrasts, which compare one treated arm against another rather than against control.

silence; the one comparison of interruption against no interruption is $p = 0.15$, on one model, and was never rechecked, so any reading of an arm’s absolute cost against an uninterrupted agent is unbounded.

7. What this means

For practitioners the inference is the reverse of the popular one. Manners are not the lever; demand is, and the work it buys is mostly repeated. The hazard runs the other way: a pleantry to an agent that is still working reads as a closing move and curtails it. For the literature, the claim is narrower and harder to dispute than “these results are wrong” — it is that **the factor these papers vary is not the factor they name**, and that an agentic setting with verifiable ground truth is where the difference becomes visible.

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